

Template Week 3 – Hardware

Student number: 530010

Assignment 3.1: Examine your phone

What processor is in your phone?

Octa-core (4x2.4 GHz Cortex-A78 & 4x1.8 GHz Cortex-A55)

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

RISC

How much RAM is in it?

6 GB RAM

How much storage does your phone have?

128 GB

What operating system is running on your phone?

Android-versie 16, Versie One UI 8.0

Approximately how many applications do you have installed?

142

Which application do you use the most?

TikTok en Opera

Can your phone be charged with what type of plug?

USB Type-C 2.0

Which I/O ports can you visually see on your phone?

Alleen de USB port

Assignment 3.2: Examine your laptop

What processor is in your laptop?

11th Gen Intel(R) Core(TM) i7-11800H @ 2.30GHz (2.30 GHz)

To which architecture family does this processor belong? In other words, which Instruction Set Architecture (ISA) is used?

RISC

How much RAM is in it?

16 GB RAM

How much storage does your laptop have?

1 TB

Which operating system is running on your laptop?

Windows 11

Approximately how many applications do you have installed?

127

Which application do you use the most?

Opera, IntelliJ en discord

Can your laptop be charged with what type of plug?

Connector: 4.5mm x 3.0mm

Which I/O ports can you visually see on your laptop?

HDMI, USB, Ethernet, port voor oplader en oortjes

Assignment 3.3: Power to the laptop

What is the input voltage?

AC-ingang (100-240V, 50/60Hz)

What is the output voltage?

DC-uitgang van 20V bij 9A (180W)

How many watts can your power adapter deliver?

180W

Is the input voltage AC or DC?

AC

Is the output voltage AC or DC?

DC

AC/DC what is that?

EEN BAND(AC/DC). AC/DC verwijst naar twee soorten electriciteit: AC (Alternating Current/wisselstroom), de stroom uit het stopcontact die constant van richting wisselt en DC (Direct Current/gelijkstroom), stroom met een constante richting

If you reverse the polarity of the output voltage, is that bad for your laptop?

Yes probably

You forgot your power adapter, your laptop normally needs 15 watts. You will be loaned a power adapter that can deliver 50 watts. Voltage, polarity, etc. are all the same compared to the original power adapter. You can connect the borrowed power adapter to your laptop. What will happen? Also explain why you think that.

Je laptop zal gewoon opladen zonder problemen

Assignment 3.4: Build your dream PC

Screenshots PC configuration + motivation:

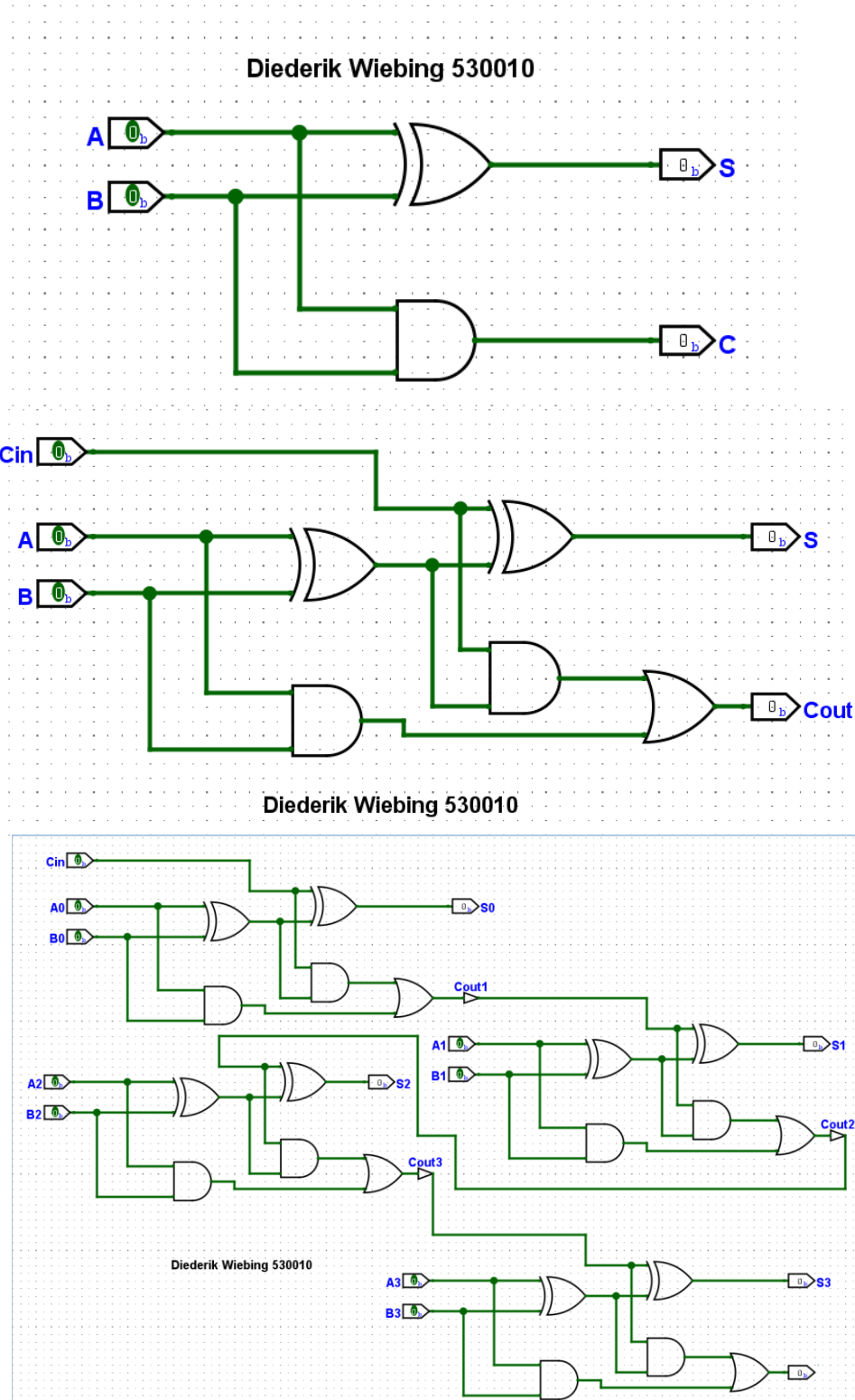
Component	Selection	Base	Promo	Shipping	Tax	Availability	Price	Where	
CPU	 AMD Ryzen 7 9800X3D 4.7 GHz 8-Core Processor	€469.00	—	FREE	—	In stock	€469.00	 AZERTY	Buy ✕
CPU Cooler	 ARCTIC Liquid Freezer III Pro A-RGB 360 77 CFM Liquid CPU Cooler	€94.49	—	 Prime	—	In stock	€94.49	 amazon.nl	Buy ✕
Motherboard	 MSI MAG B850 TOMAHAWK MAX WIFI ATX AM5 Motherboard	€253.99	—	 Prime	—	In stock	€253.99	 amazon.nl	Buy ✕
Memory	 Crucial Pro Overclocking 32 GB (2 x 16 GB) DDR5-6000 CL36 Memory	€406.99	—	 Prime	—	In stock	€406.99	 amazon.nl	Buy ✕
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	+ Add Additional Memory								
Storage	 Samsung 9100 PRO 8 TB M.2-2280 PCIe 5.0 X4 NVME Solid State Drive	€1082.90	—	FREE	—		€1082.90	 PARADIGM	Buy ✕
	+ Add Additional Storage								
Video Card	 NVIDIA Founders Edition GeForce RTX 5090 32 GB Video Card	€3289.99	—	—	—	In stock	€3289.99	 amazon.nl	Buy ✕
	+ Add Another Video Card								
Case	 Corsair 3500X ARGB ATX Mid Tower Case	€129.00	—	FREE	—	In stock	€129.00	 ALTERNATE	Buy ✕
Power Supply	 Asus ROG STRIX 1200P Gaming 1200 W 80+ Platinum Certified Fully Modular ATX Power Supply	€232.55	—	 Prime	—	In stock	€232.55	 amazon.nl	Buy ✕
Operating System	 Microsoft Windows 11 Pro OEM - DVD 64-bit	€157.76	—	FREE	—	In stock	€157.76	 amazon.nl	Buy ✕
Monitor	 Asus ROG Swift OLED PG27AQDP 26.5" 2560 x 1440 480 Hz Monitor	€987.00	—	 Prime	—	In stock	€987.00	 amazon.nl	Buy ✕
	+ Add Another Monitor								

Beste van het beste zodat ik mooi alle games kan spelen die ik wil zonder probleem

Assignment 3.5: Adders

Complete the **half adder**, **full adder** and **4-bit adder** assignment as described in the PowerPoint slides of week 3 in Logisim. Save the chip design and also export three PNG pictures of the separate finished designs. See the PowerPoint slides of week 3.

Paste the three exported PNG pictures in here.



Ready? Save this file and export it as a pdf file with the name: [week3.pdf](#)